

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit: Mining Engineering**
2. **Subject Name: Mine Computational Laboratory L-T-P: 0-0-3 Credits:2**
3. **Pre-requisites: NIL**
4. **Syllabus and reference books:**

Syllabus: The course includes experiments from all subject areas, which the students have studied individually. This laboratory aimed to assimilate all knowledge to solve the industrial problems of mining and allied industries. The experiments will be designed from (1) Mine systems, (2) selection of equipment for safe and economical operations, (3) ventilation planning, (4) Design of underground and surface excavation, (5) Environmental impact assessment and control, (6) Manpower planning and assessment services for new mining projects, (7) Assessment of Mineral processing and coal beneficiation plant and others.

Reference Books:

1. Misra G. B. (1986). Mine Environment and Ventilation, OUP.
2. Hustrulid, W. A., Kuchta, M., & Martin, R. K. (2013). Open pit mine planning and design, two volume set & CD-ROM pack. CRC Press.
3. De A. (2014). Latest Development of Heavy Earth Moving Machinery. Galgotias Publisher.
4. Walker S. C. (1988). Mine Winding and Transport. Elsevier.
5. Ramlu M. A. (2015). Mine Hoisting, White falcon Publishing.
6. Wills Barry A., Finch James (2015) An Introduction to the Practical Aspects of Ore Treatment and Mineral Recovery, Wills' Mineral Processing Technology, 8th Edition.
7. Obert L., Duvall W. I., (1967) Rock Mechanics and the Design of Structures, John Wiley & Sons Inc.

5. List of Experiments:

SN	Experiment Name
1.	Study of underground and surface mine plans
2.	Preparation of Feasibility and pre-feasibility report
3.	Selection of mine machinery and design of underground opening based on mine production.
4.	Selection of mine machinery for various operations in opencast mining
5.	Ventilation network design and selection of ventilation equipment for an underground mine
6.	Design of underground opening design
7.	Design of surface mine design
8.	Assessment of mineral processing plant or coal washeries for a mining operation
9.	Manpower management and assessment of services for a new mining project.
10.	Environment impact assessment of a mining project and its control measures